llnl_c1/lulesh · lulesh.cc · llm_pragmas_v5 · claude-haiku-4-5-20251001 · BROKEN · RIGHT = attempt 1 of 41, rejected by the gate at 'tsan'

−5 / +26 lines of 1967 · 5 parallel region(s) added · 17 of 41 candidate(s) kept · RESTRUCTURING (11 code line(s) added, 3 removed) · restructuring level 0

BEFORE

```
1953  {
1954  #ifdef SEDOV_SYNC_POS_VEL_LATE
1955     Domain_member fieldData[6] ;
1956  #endif
1957
1958     LagrangeNodal(domain);
1959
1960  #ifdef SEDOV_SYNC_POS_VEL_LATE
1961  #endif
1962
1963     LagrangeElements(domain, domain.numElem());
1964
1965     CalcTimeConstraintsForElems(domain);
1966
1967  }
```

ATTEMPT 1 OF 41, REJECTED BY THE GATE AT 'TSAN'

```
1953  {
1954  #ifdef SEDOV_SYNC_POS_VEL_LATE
1955     Domain_member fieldData[6] ;
1956  #endif
1957
1958  #pragma omp parallel shared(domain)
1959  {
1960     /* Three independent function calls with no data dependences;
1961        each operates on disjoint domain data and can execute in parallel */
1962
1963  #pragma omp task
1964     {
1965        /* Update nodal properties: forces, accelerations, velocities, positions
1966        LagrangeNodal(domain);
1967     }
1968
1969  #pragma omp task
1970     {
1971        /* Update element properties based on current node data */
1972        LagrangeElements(domain, domain.numElem());
1973     }
1974
1975  #pragma omp task
1976     {
1977        /* Compute time constraints for next time step */
1978        CalcTimeConstraintsForElems(domain);
1979     }
1980
1981     /* Wait for all three tasks to complete */
1982  #pragma omp taskwait
1983  }
1984
1985  #ifdef SEDOV_SYNC_POS_VEL_LATE
1986  #endif
1987


1988  }
```

file

red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is